

Fundamentals of Electric Circuits

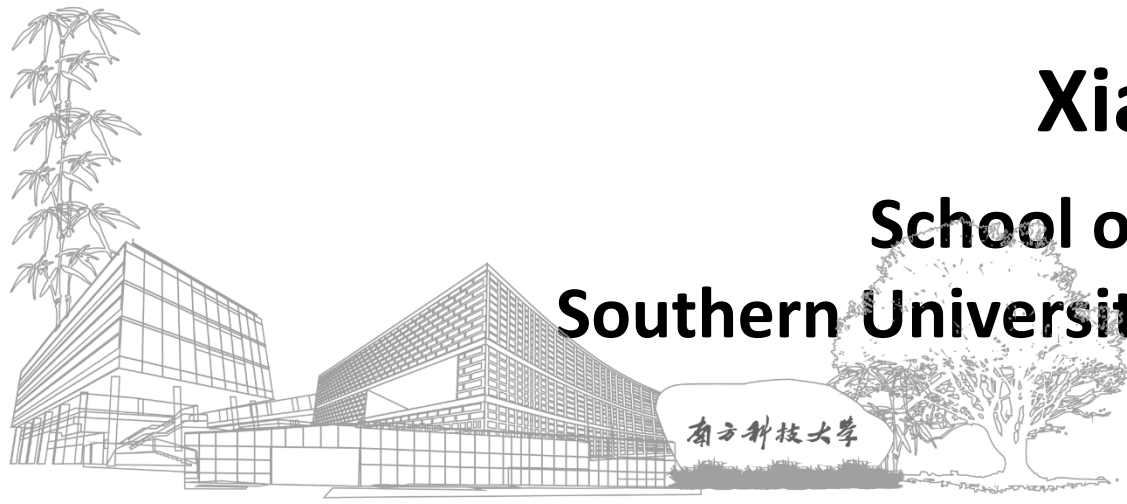
电路基础

Lecture8: Second-Order Circuits

Xiaolong Liu

School of Microelectronics

Southern University of Science and Technology

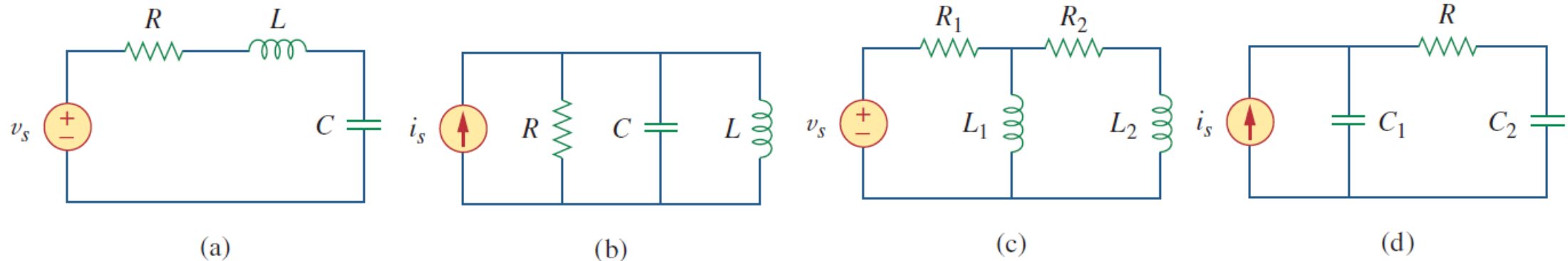


Outline

- **Introduction**
- **Finding Initial and Final Values**
- **The Source-Free Series *RLC* Circuit**
- **The Source-Free Parallel *RLC* Circuit**
- **Step Response of a Series *RLC* Circuit**
- **Step Response of a Parallel *RLC* Circuit**
- **General Second-Order Circuits**

Introduction

- In this lecture, we consider **second-order circuits (二阶电路)**, i.e., circuits that **contain two storage elements** and can be described by **second-order differential equations**
- Some typical second-order circuits are shown below. Note that the two storage elements can be of different type (such as in (a) and (b)) or same type (such as in (c) and (d), where the elements cannot be represented by an equivalent single element)



- The circuits can be excited by initial conditions of the storage elements and by independent sources. We will examine their reactions in both the **source-free** and **switched (dc) source** scenarios

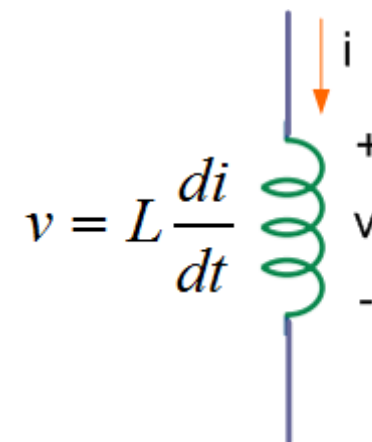
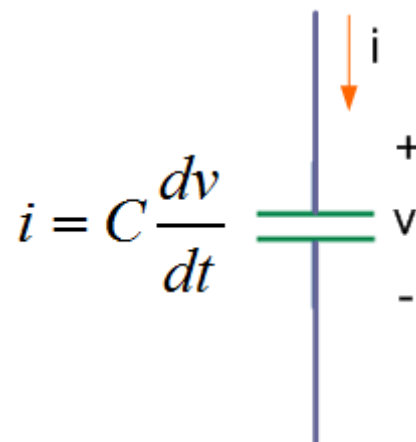
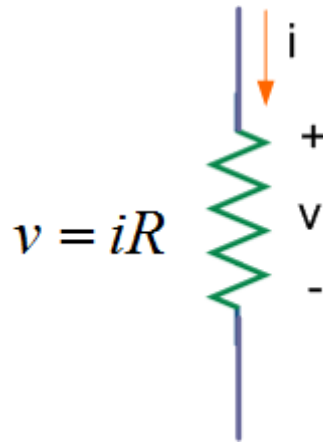
Finding Initial and Final Values (I)

- Working on second-order system is harder than first-order in terms of finding initial and final conditions (初值和终值)
- The initial-value problem (IVP, 初值问题) for a second-order differential equation requires information on the initial conditions of the form $y(x_0) = y_0$ and $y'(x_0) = y_1$. For second-order circuits, capacitor voltage v and inductor current i are the primary functions we would like to find → Finding $v(0)$, $i(0)$, $dv(0)/dt$, and $di(0)/dt$
- When we consider the systematic approaches of decomposing the step responses into steady-state and transient ones, we would also need to know the steady-state or final values, i.e., $v(\infty)$ and $i(\infty)$ → It turns out that finding initial and final values, i.e., $v(0)$, $i(0)$, $v(\infty)$ and $i(\infty)$ are relatively easy; while finding the initial values of their derivatives, i.e., $dv(0)/dt$, and $di(0)/dt$ are not that straightforward



Finding Initial and Final Values (II)

- Two key points to keep in mind in determining the initial conditions
 - 1). As always in circuit analysis, we must carefully handle the polarity of voltage across a capacitor and the direction of current through an inductor. Keep in mind that v and i are defined strictly according to the **passive sign convention**
 - 2). Capacitor voltage and inductor current **always remain continuous** $\rightarrow v(0^+) = v(0^-)$ for capacitor, and $i(0^+) = i(0^-)$ for inductor, where $t = 0^+$, 0^- denotes the time just before and after the starting instant that switching, or application of a power source occurs



Example: Finding Initial and Final Values

- The switch in the figure has been closed for a long time. It is opened at $t = 0$. Find: (a) $i(0^+)$, $v(0^+)$, (b) $di(0^+)/dt$, $dv(0^+)/dt$, (c) $i(\infty)$, $v(\infty)$

Solution:

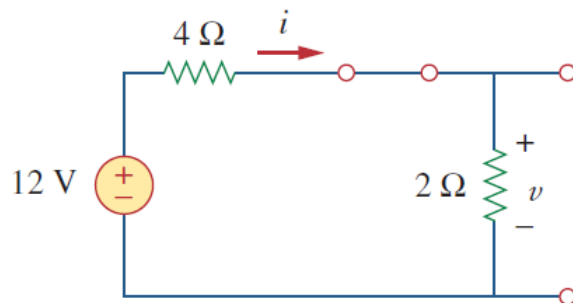
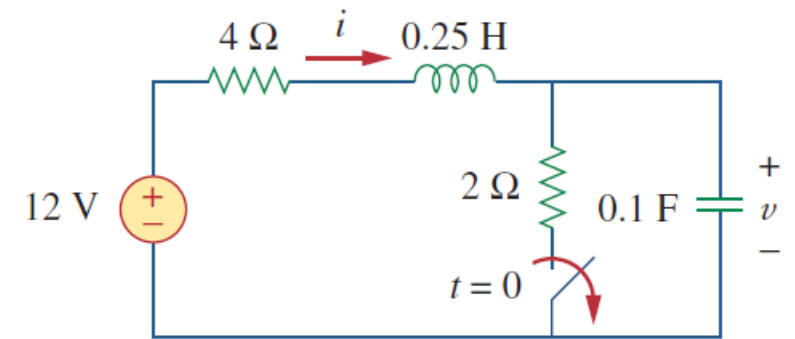
The equivalent circuit for $t = 0^-$ and $t = 0^+$ and $t \rightarrow \infty$ are shown in Fig. (a), (b) and (c), respectively

(a) The switch has been closed for a long time before $t = 0$, so dc steady state has been reached. Therefore, $i(0^+) = i(0^-) = 2$ A; $v(0^+) = v(0^-) = 12 \cdot 2 / (2 + 4) = 4$ V

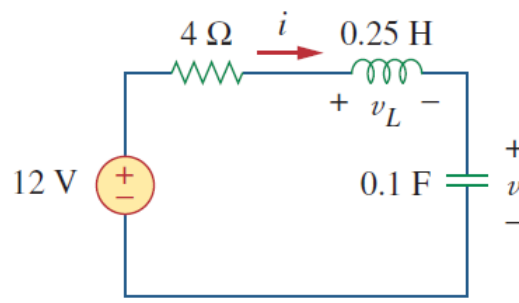
(b) According to KVL, $v_L(0^+) = 12 - 4i(0^+) - v(0^+) = 0$ V, so $di(0^+)/dt = v_L(0^+)/L = 0$ A/s

Since $i(0^+) = C \cdot dv(0^+)/dt$, $dv(0^+)/dt = i(0^+)/C = 2/0.1 = 20$ V/s

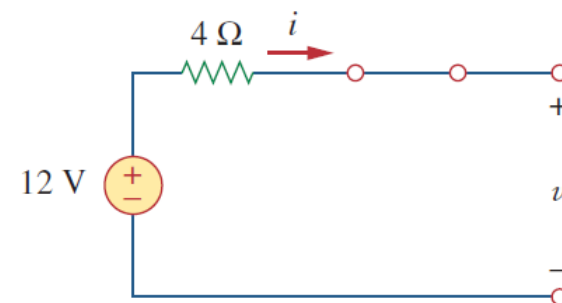
(c) As $t \rightarrow \infty$, the circuit reaches steady state again, just with the switch opened. So, $i(\infty) = 0$ A; $v(\infty) = 12$ V



(a)



(b)



(c)

The Source-Free Series RLC Circuit (I)

- An understanding of the **natural response** of the series RLC circuit is a necessary background for future studies in filter design and communications networks
- Consider the series RLC circuit shown below. The energy at $t = 0$ is stored in the capacitor, represented by V_0 and in the inductor, represented by I_0
- Applying KVL around the loop, we have

$$Ri + L\frac{di}{dt} + \frac{1}{C} \int_{-\infty}^t i(\tau)d\tau = 0 \quad \Rightarrow \quad \boxed{\frac{d^2i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0}$$

- This is a homogeneous second-order differential equation. Its initial conditions are

$$\boxed{\begin{cases} i(0) = I_0 \\ \frac{di(0)}{dt} = \frac{0 - RI_0 - V_0}{L} = -\frac{1}{L}(RI_0 + V_0) \end{cases}}$$

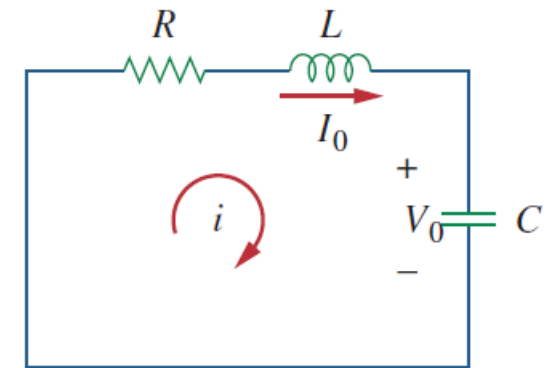


Figure 8.8

A source-free series RLC circuit.

Revisit: Second-Order Differential Equation

- It is always possible to discover **particular solutions** (特解) to a second-order linear equation if the coefficient functions P, Q, and R are constant functions

$$\boxed{\frac{dy}{dx} + P(x)y = Q(x)} \quad \rightarrow \quad \boxed{ay'' + by' + cy = 0}$$

- Where a, b, and c are constants and $a \neq 0$. We will see constantly that this is the case in this course, for the linear and time-invariant circuits
- If $y = e^{rx}$ is a solution, then we have

$$\boxed{ar^2 e^{rx} + bre^{rx} + ce^{rx} = 0 \rightarrow (ar^2 + br + c)e^{rx} = 0 \rightarrow ar^2 + br + c = 0}$$

- The equation $ar^2 + br + c = 0$ is called the **characteristic equation** (特征方程) of the differential equation $ay'' + by' + cy = 0$. We can use quadratic formula to find its roots

$$\boxed{ar^2 + br + c = 0 \rightarrow r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}}$$

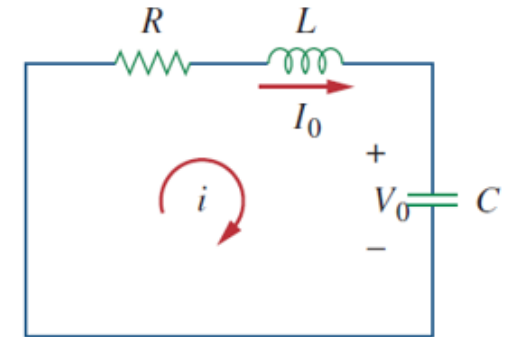
The Source-Free Series *RLC* Circuit (II)

- The **general solution** of a homogeneous 2nd-order differential equation $ay''+by'+cy=0$ takes **three different forms** according to the **discriminant** (判别式) $b^2 - 4ac$ of its **characteristic equation** (特征方程)
- For the *RLC* circuit, $a = 1$, $b = R/L$, and $c = 1/LC$. The characteristic equation is

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

- Using quadratic formula, the two roots of the characteristic equation are

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}, s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$



$$\frac{d^2i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

- The roots s_1 and s_2 are called **natural frequencies** (自然频率), measure in nepers per second (Np/s), since they are associated with the **natural response** of the circuit

The Source-Free Series *RLC* Circuit (III)

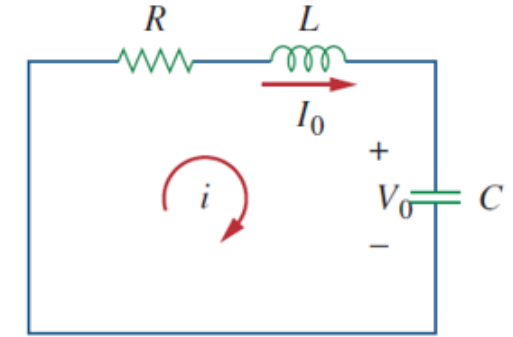
- Using compact ways of expression, the two roots are given by

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

where $\alpha = \frac{R}{2L}$ and $\omega_0 = \frac{1}{\sqrt{LC}}$

- Parameter α is called **damping factor** (阻尼系数、衰减系数), and ω_0 is **resonant frequency** (自由振荡角频率) or **undamped natural frequency**
- In terms of α and ω_0 , the characteristic equation can also be written as

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 \quad \Rightarrow \quad s^2 + 2\alpha s + \omega_0^2 = 0$$



$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

- The general solution of the homogeneous second-order differential equation can be classified into three cases, according to the values of the discriminant $\alpha^2 - \omega_0^2$, which is determined by the relative value of α and ω_0

The Source-Free Series *RLC* Circuit (IV)

- If $\alpha > \omega_0$, the roots s_1 and s_2 are real and distinct, both negative. The general solution is given by

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

- It decays and approaches zero as t increases. This is the **overdamped (过阻尼)** case

- If $\alpha = \omega_0$, $s_1 = s_2 = s = -\alpha$. The general solution is given by

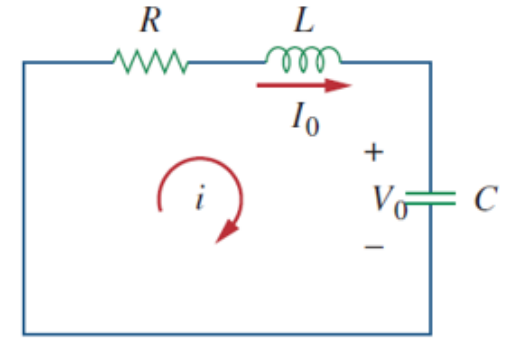
$$i(t) = e^{-\alpha t} (A_1 + A_2 t)$$

- This is the **critically damped (临界阻尼)** case

- If $\alpha < \omega_0$, $s_1 = -\alpha + \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha + j\omega_d$, $s_2 = -\alpha - \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha - j\omega_d$. The general solution is given by

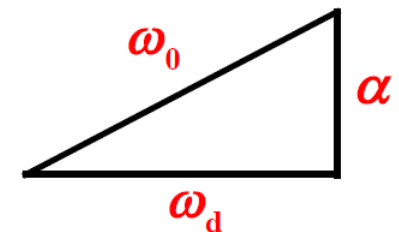
$$i(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t), \text{ where } \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

- Where parameter ω_d is called the **damped natural frequency (阻尼/衰减振荡角频率)**. This is the **underdamped (欠阻尼)** case



$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

$$s^2 + 2\alpha s + \omega_0^2 = 0$$

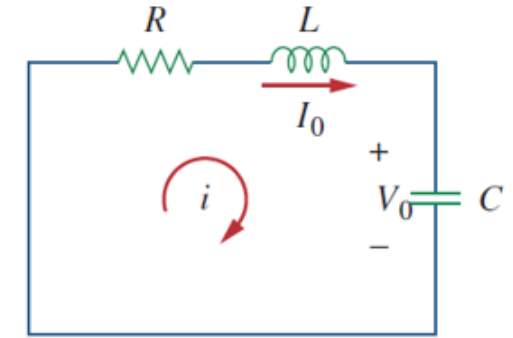


The Source-Free Series *RLC* Circuit (V)

- If $R = 0$, hence $\alpha = 0$, we have the **undamped (无阻尼)** case, and the solution is

$$i(t) = (A_1 \cos \omega_0 t + A_2 \sin \omega_0 t)$$

- Which shows that the current maintains an oscillation at the resonant frequency



$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

$$\alpha = \frac{R}{2L} \text{ and } \omega_0 = \frac{1}{\sqrt{LC}}$$

Example:

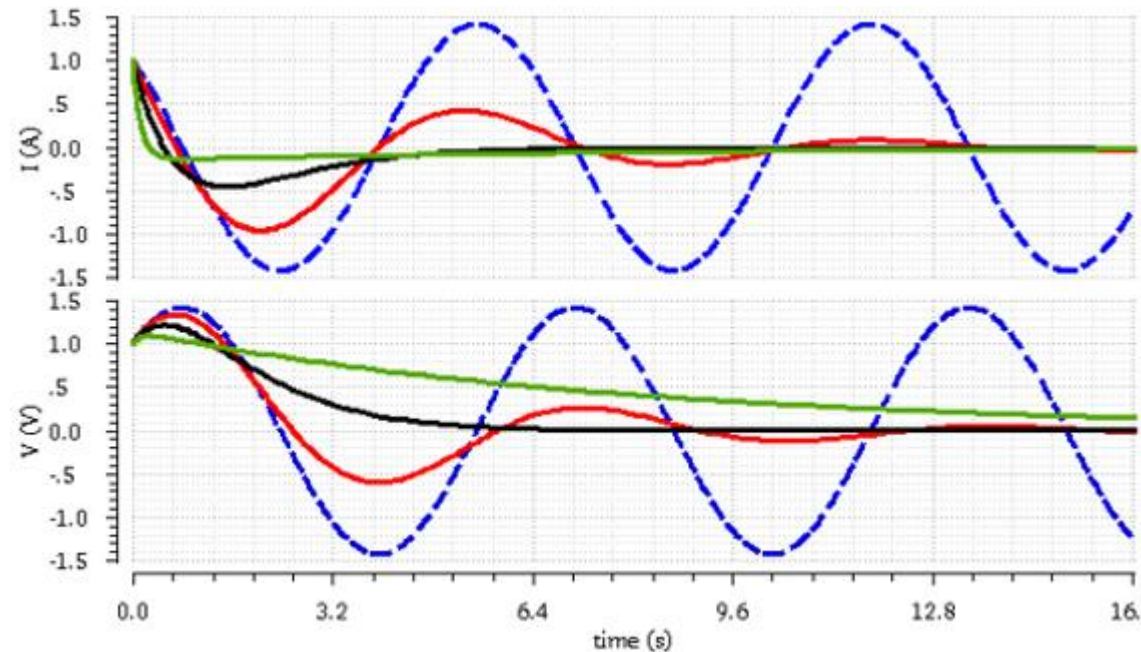
$L = 1 \text{ H}$, $C = 1 \text{ F}$, $I_0 = 1 \text{ A}$, $V_0 = 1 \text{ V}$

$R = 0 \Omega$, $\alpha = 0$, **undamped.**

$R = 0.5 \Omega$, $\alpha = 0.25$, **underdamped**

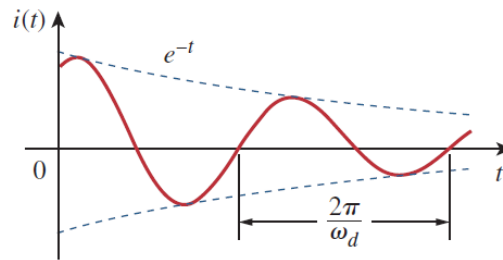
$R = 2 \Omega$, $\alpha = 1$, **critically damped**

$R = 8 \Omega$, $\alpha = 4$, **overdamped**



Summary of The Behaviors of Series *RLC* Circuit

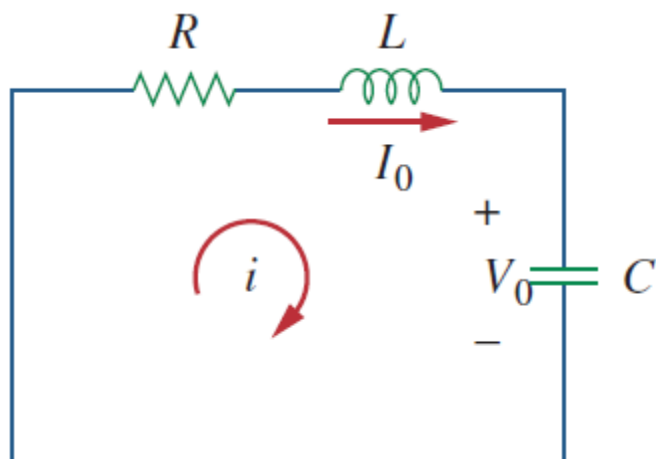
- The *RLC* networks can be characterized by the following:
- The behavior of such a network is captured by the idea of **damping (阻尼)**, which is the **gradual loss of the initial stored energy**, due to the presence of resistance R . The damping factor determines the rate at which the response is damped. By adjusting the value of R , the response may be made **undamped, overdamped, critically damped, or underdamped**
- Oscillatory response** is possible due to the presence of two types of energy storage elements. Having both L and C allows the flow of energy back and forth between the two. The damped oscillation exhibited by the underdamped response is known as **ringing (振铃)**



- It is typically difficult to tell the difference between **damped and critically damped responses (过阻尼与临界阻尼响应较难区分)**

Example1: The Source-Free Series *RLC* Circuit

- In the following figure, $R = 40 \, \Omega$, $L = 4 \, \text{H}$, and $C = 1/4 \, \text{F}$. Calculate the characteristic roots of the circuit. Is the natural response overdamped, underdamped, or critically damped?



$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

$$s^2 + \frac{R}{L} s + \frac{1}{LC} = 0$$

Solution:

We can find the damping factor α and resonant frequency ω_0 first.

$$\alpha = \frac{R}{2L} = \frac{40}{2 \cdot 4} = 5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \cdot \frac{1}{4}}} = 1$$

The roots are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -5 \pm \sqrt{25 - 1}$$

$$\rightarrow s_1 = -0.101, s_2 = -9.899$$

Since $\alpha > \omega_0$, the natural response is overdamped

Example2: The Source-Free Series *RLC* Circuit

- Find $i(t)$ for $t > 0$ in the circuit on the right. Assume that the circuit has reached steady state at $t = 0^-$

Solution:

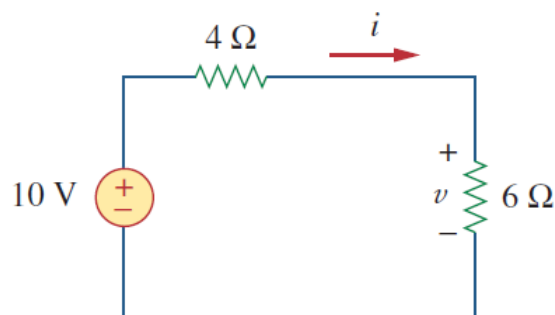
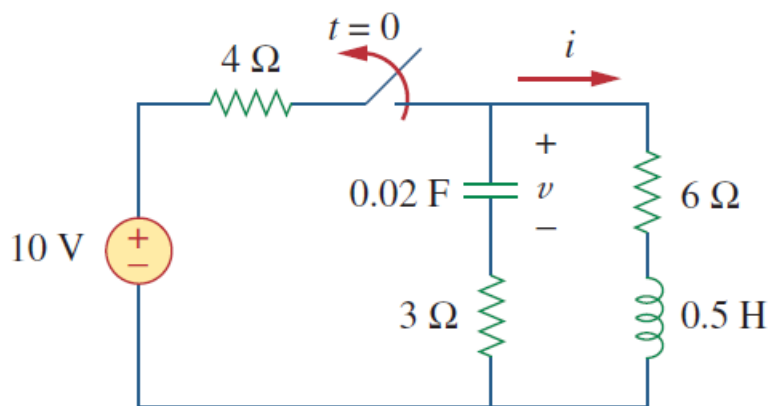
After the opening of switch at $t = 0$, the right part of the circuit becomes a source-free **series *RLC***, in which $R = 3 + 6 = 9\ \Omega$, $L = 0.5\ \text{H}$, $C = 0.02\ \text{F}$.

The damping factor $\alpha = R/2L = 9\ \text{rad/s}$, the natural frequency $\omega_0 = 1/\sqrt{L \times C} = 10\ \text{rad/s}$. Now that $\alpha < \omega_0$, the circuit is **underdamped**. Its general solution for $t > 0$ is hence given by

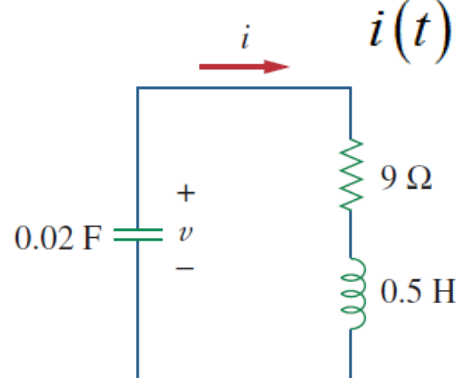
$$i(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t) \quad \text{Where } \omega_d = \sqrt{\omega_0^2 - \alpha^2} = 4.359$$

Consider the initial conditions: $v(0^+) = 10 \times 6/(4 + 6) = 6\ \text{V}$;
 $v_L(0^+) = v(0^+) - i(0^+) \times (3 + 6) = -3\ \text{V}$. therefore, we have
 $i'(0^+) = v_L(0^+)/L = -6\ \text{A/s}$ and $i(0^+) = 10/(4 + 6) = 1\ \text{A}$
 Substituting these values into $i(t)$, we have $A_1 = 1$ and $A_2 = 0.6882$

$$i(t) = e^{-9t} (\cos 4.359t + 0.6882 \sin 4.359t)$$



(a)



(b)

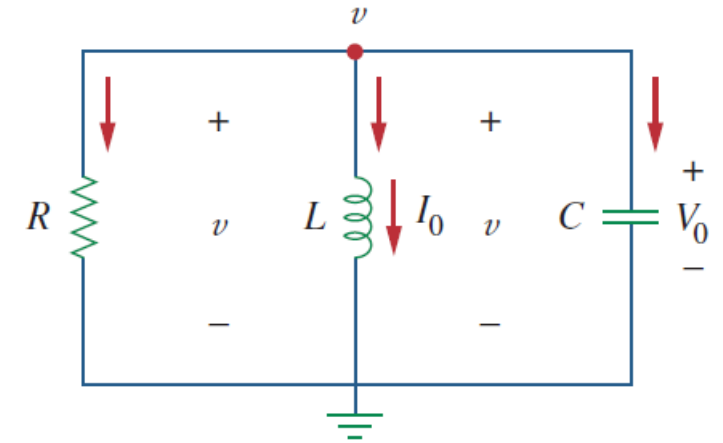
The Source-Free Parallel RLC Circuit (I)

- Now consider the parallel form of RLC circuit, as shown below. The initial stored energy in capacitor and inductor at $t = 0$ are specified by V_0 and I_0
- Applying KCL at the top node, we have

$$\frac{v}{R} + \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau + C \frac{dv}{dt} = 0 \Rightarrow \frac{d^2 v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

- With initial conditions

$$\begin{cases} v(0) = V_0 \\ \frac{dv(0)}{dt} = \frac{0 - V_0/R - I_0}{C} = -\frac{1}{RC}(V_0 + RI_0) \end{cases}$$



- The characteristic equation and its roots are given by

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0 \Rightarrow s_{1,2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \Rightarrow$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

where $\alpha = \frac{1}{2RC}$ and $\omega_0 = \frac{1}{\sqrt{LC}}$

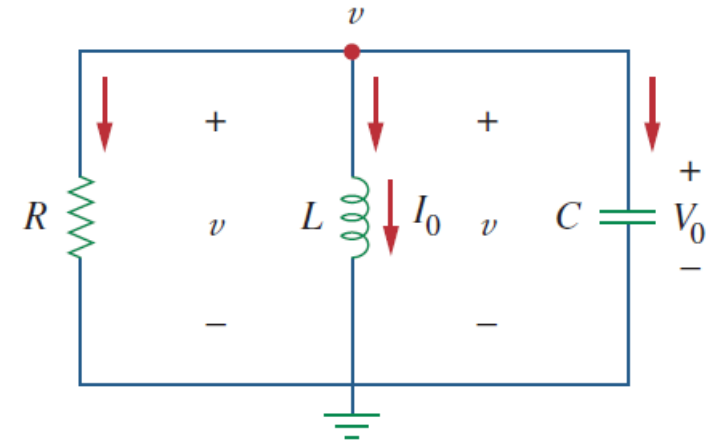
- The general solution of the homogeneous second-order differential equation can be classified into three cases, according to the values of the discriminant $\alpha^2 - \omega_0^2$, which is determined by the relative value of α and ω_0

The Source-Free Parallel *RLC* Circuit (II)

- The same as the case of series RLC, the response can be classified into three categories, depending on their damping or oscillating behaviors
- **Overdamped (过阻尼):** $\alpha > \omega_0$, the two roots s_1 and s_2 are different negative real numbers. The solution $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ dies off to zero with t
- **Critically Damped (临界阻尼):** $\alpha = \omega_0$, the two roots are identical and have a negative real value, $s_1 = s_2 = -\alpha$. The solution $v(t) = e^{-\alpha t} (A_1 + A_2 t)$
- **Underdamped (欠阻尼):** $\alpha < \omega_0$, the two roots are different complex number with negative real part, $s_1 = -\alpha + j\omega_d$ and $s_2 = -\alpha - j\omega_d$. The solution $v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ is called **damped natural frequency (阻尼频率)**
- **Undamped (无阻尼):** $\alpha = 0$, the two roots are different imaginary numbers, $s_1 = j\omega_d$ and $s_2 = -j\omega_d$. The solution $v(t) = A_1 \cos \omega_0 t + A_2 \sin \omega_0 t$

Example: The Source-Free Parallel *RLC* Circuit (I)

- In the parallel circuit on the right, find $v(t)$ for $t > 0$, assuming $v(0) = 5 \text{ V}$, $i(0) = 0$, $L = 1 \text{ H}$, and $C = 10 \text{ mF}$. Consider these cases: $R = 1.923 \Omega$, $R = 5 \Omega$, and $R = 6.25 \Omega$



Solution:

The initial conditions are:

$$v(0) = V_0 = 5 \text{ V, and}$$

$$v'(0) = -(V_0 + R \times I_0)/RC = -260, -100, -80 \text{ V/s for } R = 1.923\Omega, 5\Omega, \text{ and } 6.25 \Omega, \text{ respectively.}$$

$$\text{The resonant frequency } \omega_0 = 1/\sqrt{LC} = 10 \text{ rad/s}$$

Case 1: $R=1.923 \Omega$

$$\text{The damping factor } \alpha = 1/2RC = 1/(2 \times 1.923 \times 10e^{-3}) = 26 > \omega_0$$

This is an **overdamped case**, and the response for $t > 0$

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{where } s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -2, s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -50$$

Example: The Source-Free Parallel *RLC* Circuit (II)

Substituting the initial conditions in $v(t)$

$$\begin{cases} 5 = (A_1 \cdot 1 + A_2 \cdot 1) \\ -260 = A_1 \cdot s_1 \cdot 1 + A_2 \cdot s_2 \cdot 1 = -2A_1 - 50A_2 \end{cases} \rightarrow A_1 = -0.2083, A_2 = 5.2083$$

The voltage across the capacitor is $v(t) = -0.2083e^{-2t} + 5.2083e^{-50t}$

Case 2: $R = 5 \Omega$:

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 5 \times 10 \times 10^{-3}} = 10 = \omega_0$$

so, the natural response is **critically damped** and $v(t) = e^{-\alpha t} (A_1 + A_2 t)$

Using the initial conditions, we get

$$\begin{cases} 5 = 1 \cdot (A_1 \cdot 1 + A_2 \cdot 0) \\ -100 = -10 \cdot (A_1 \cdot 1 + A_2 \cdot 0) + 1 \cdot (0 + A_2) = -10A_1 + A_2 \end{cases} \rightarrow \boxed{v(t) = e^{-10t} (5 - 50t)}$$

$$\rightarrow A_1 = 5, A_2 = -50$$

Example1: Source-Free Parallel *RLC* Circuit (II)

Case 3: $R = 6.25 \Omega$:

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 6.25 \times 10 \times 10^{-3}} = 8 < \omega_0$$

so, the natural response is **underdamped** and

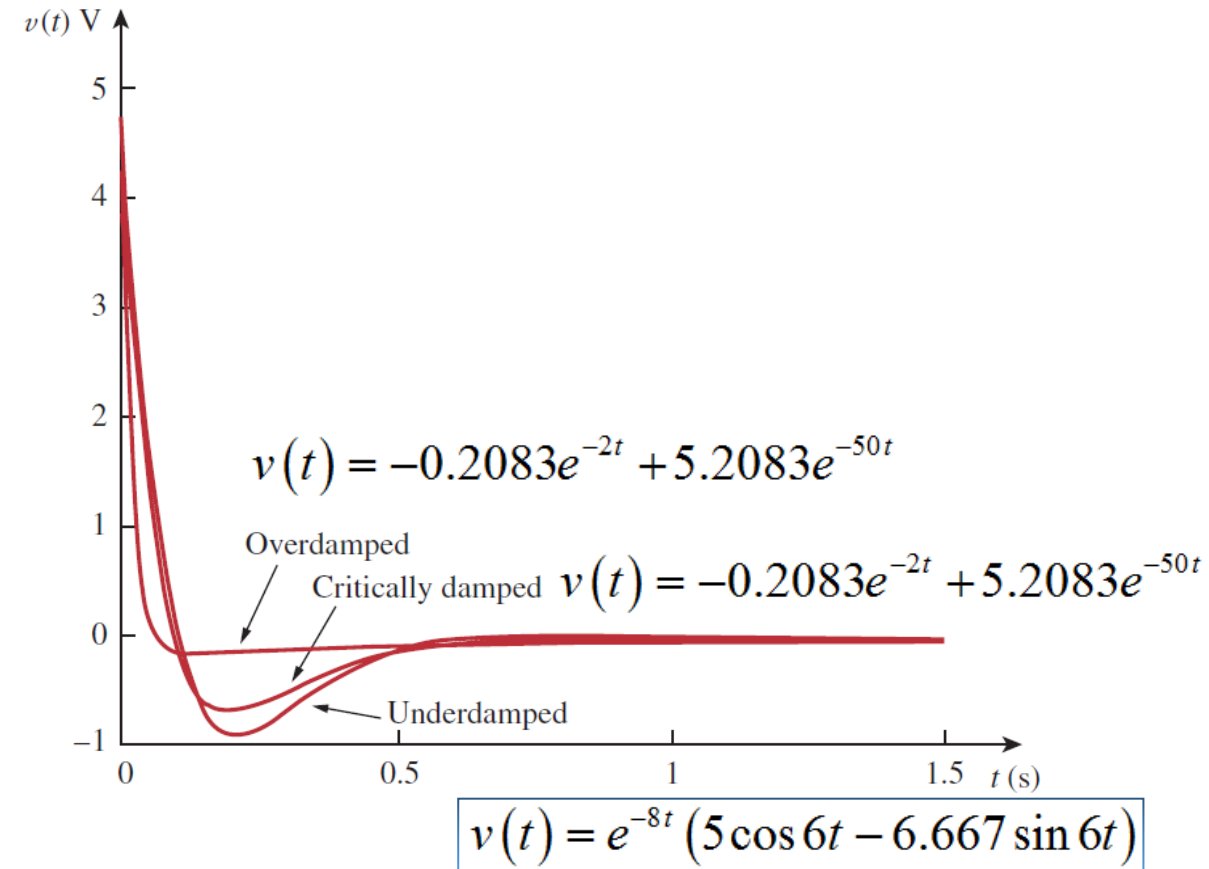
$$v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

Using the initial conditions, we get

$$\begin{cases} 5 = 1 \cdot (A_1 \cdot 1 + A_2 \cdot 0) \\ -80 = 1 \cdot (-6A_1 \cdot 0 + 6A_2 \cdot 1 - 8A_1 \cdot 1 - 8A_2 \cdot 0) = -8A_1 + 6A_2 \end{cases}$$

$$\rightarrow A_1 = 5, A_2 = -6.667$$

Therefore, $v(t) = e^{-8t} (5 \cos 6t - 6.667 \sin 6t)$



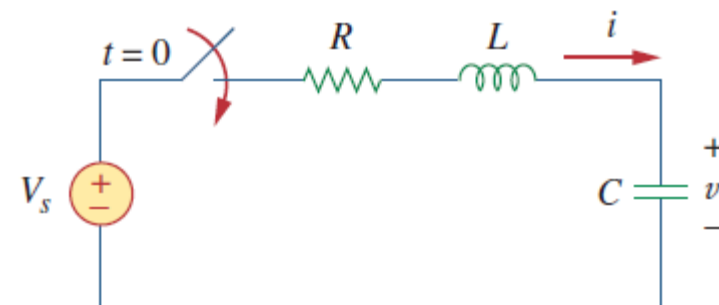
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- The Source-Free Parallel *RLC* Circuit
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- **General Second-Order Circuits**

Step Response of a Series *RLC* Circuit (I)

- Consider the series *RLC* circuit shown right. Applying KVL, we have (for $t > 0$)

$$L \frac{di}{dt} + Ri + v = V_s \quad \xrightarrow{i = C \frac{dv}{dt}} \quad \boxed{\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}}$$



- This's a nonhomogeneous second-order differential equation. Its **complementary equation** (补余方程) is the same as in the source-free series *RLC* circuit
- For the **particular solution** (特解), we can easily find that a particular solution is $v = V_s$. The general solution is then given by

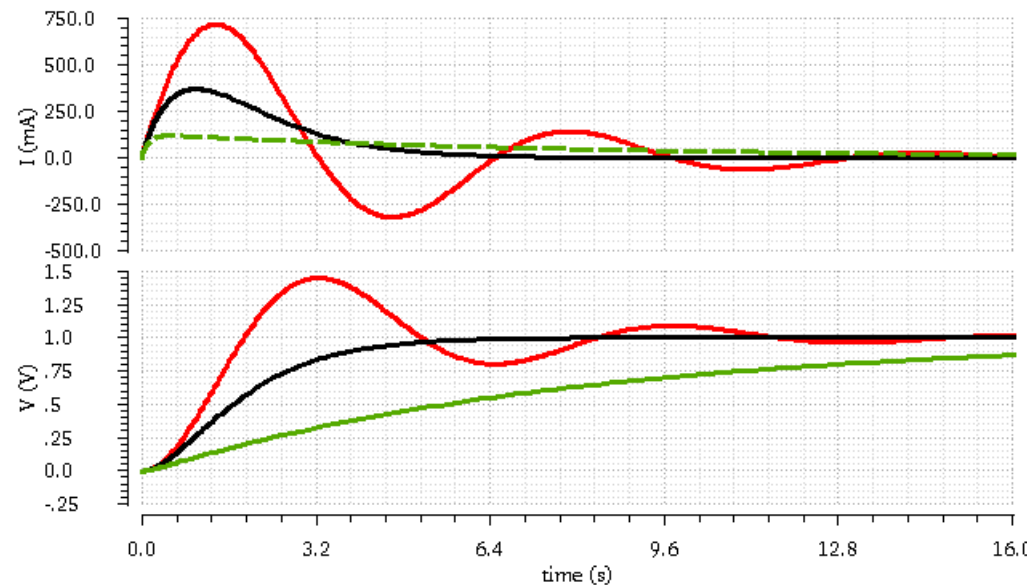
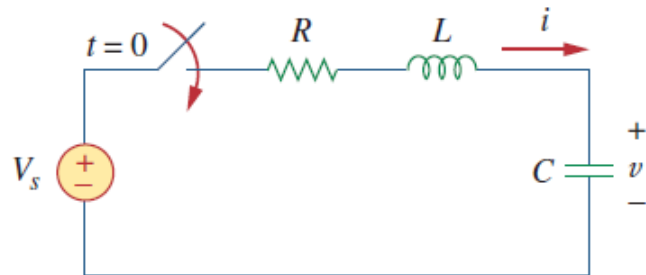
$$\begin{aligned} v(t) &= V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} && \text{(Overdamped)} \\ v(t) &= V_s + (A_1 + A_2 t) e^{-\alpha t} && \text{(Critically damped)} \\ v(t) &= V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} && \text{(Underdamped)} \end{aligned}$$

Step Response of a Series *RLC* Circuit (II)

- Notice that the particular solution is also the steady-state value of the capacitor voltage: $v_{ss}(t) = v(\infty) = V_s$. The general solution can also be expressed as the sum of two parts:

$$v(t) = v_t(t) + v_{ss}(t)$$

- where $v_{ss}(t)$ is the **steady-state** part (i.e., the particular solution), while the $v_t(t)$ is the **transient part** (i.e., the part achieved by solving the complementary equation)



$$\alpha = \frac{R}{2L} \quad \text{and} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Example:

$$L = 1 \text{ H}, C = 1 \text{ F}, I_0 = 0 \text{ A}, V_0 = 0 \text{ V}$$

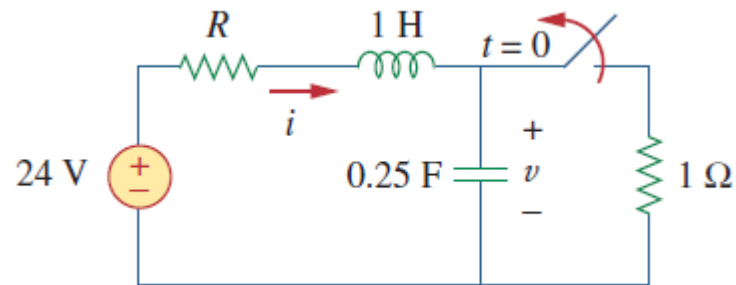
$R=0.5 \Omega$, $\alpha=0.25$, underdamped

$R=2 \Omega$, $\alpha=1$, critically damped

$R=8 \Omega$, $\alpha=4$, overdamped

Example: Step Response of a Series *RLC* Circuit (I)

- For the circuit below, find $v(t)$ for $t > 0$. Consider these cases: $R=5\ \Omega$, $R=4\ \Omega$, and $R=1\ \Omega$



Solution:

After the opening of switch at $t = 0$, the circuit becomes the series *RLC* circuit powered by a dc voltage source.

The natural resonant frequency $\omega_0 = 1/\sqrt{LC} = 2\text{ rad/s}$.

Case 1: $R = 5\ \Omega$

Initial values: $v(0^+) = 24 \times \frac{1}{R+1} = 4\text{ V}$; $v'(0^+) = I_0/C = 24/(R+1)/C = 16\text{ V/s}$.

Final value: $v(\infty) = 24\text{ V}$.

The damping factor: $\alpha = R/2L = 2.5$

Now that $\alpha > \omega_0$, the step response is **overdamped** and given by

$$v(t) = 24 + A_1 e^{s_1 t} + A_2 e^{s_2 t},$$

$$\text{where } s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -1; s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -4$$

Example: Step Response of a Series *RLC* Circuit (II)

Substituting the initial conditions, we get

$$\begin{cases} 4 = (24 + A_1 \cdot 1 + A_2 \cdot 1) \\ 16 = A_1 \cdot (-1) \cdot 1 + A_2 \cdot (-4) \cdot 1 \end{cases} \rightarrow A_1 = -\frac{64}{3}, A_2 = \frac{4}{3}$$

Finally, $v(t) = 24 - \frac{64}{3}e^{-t} + \frac{4}{3}e^{-4t}$

Case 2: $R = 4 \Omega$

Initial values: $v(0^+) = 24 \times \frac{1}{R+1} = 4.8 \text{ V}$; $v'(0^+) = 24/(R+1)/C = 19.2 \text{ V/s}$.

Final value: $v(\infty) = 24 \text{ V}$.

The damping factor: $\alpha = R/2L = 2$. Now that $\alpha = \omega_0$, the step response is critically **damped** and given by $v(t) = 24 + e^{-\alpha t}(A_1 + A_2 t)$

Using the initial conditions, we get

$$\begin{cases} 4.8 = 24 + 1 \cdot (A_1 + A_2 \cdot 0) \\ 19.2 = -2 \cdot (A_1 + A_2 \cdot 0) + 1 \cdot A_2 \end{cases} \rightarrow A_1 = -19.2, A_2 = -19.2 \Rightarrow v(t) = 24 - 19.2e^{-2t}(1+t)$$

Example: Step Response of Series *RLC* Circuit (III)

Case 3: $R = 1 \Omega$

Initial values: $v(0^+) = 24 \times \frac{1}{R+1} = 12 \text{ V}$; $v'(0^+) = 24/(R+1)/C = 48 \text{ V/s}$.

Final value: $v(\infty) = 24 \text{ V}$.

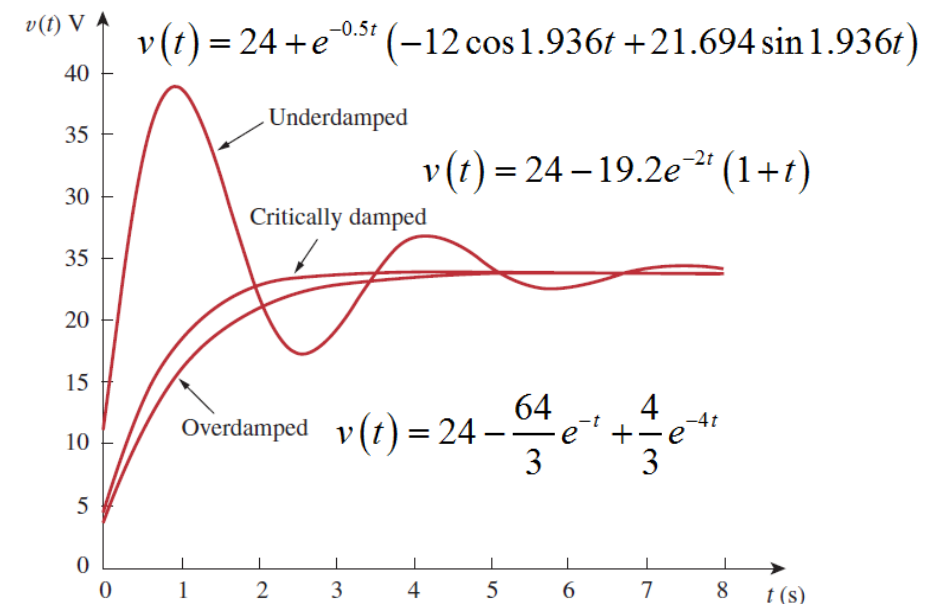
The damping factor: $\alpha = R/2L = 0.5$. Now that $\alpha < \omega_0$, the step response is **underdamped** and given

by $v(t) = 24 + e^{-0.5t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 1.936$

Substituting the initial conditions in $v(t)$, we get

$$\begin{cases} 12 = 24 + 1 \cdot (A_1 \cdot 1 + A_2 \cdot 0) \\ 48 = -0.5 \cdot (A_1 \cdot 1 + A_2 \cdot 0) + 1 \cdot (0 + 1.936 A_2 \cdot 1) \end{cases} \rightarrow A_1 = -12, A_2 = 21.694$$

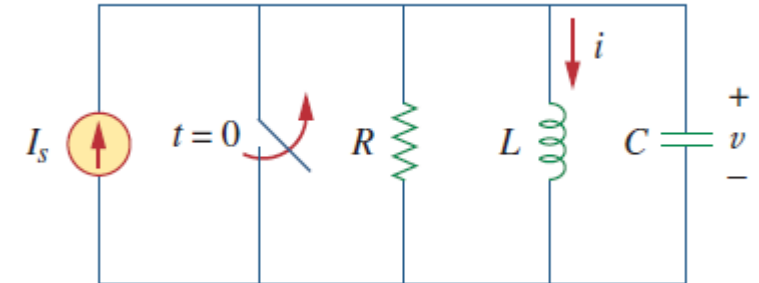
Finally, $v(t) = 24 + e^{-0.5t}(-12 \cos 1.936t + 21.694 \sin 1.936t)$



Step Response of a Parallel *RLC* Circuit (I)

- Consider the series *RLC* circuit shown right. Applying KVL, we have (for $t > 0$)

$$\frac{v}{R} + i + C \frac{dv}{dt} = I_s \quad \xrightarrow{v = L \frac{di}{dt}} \quad \boxed{\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}}$$



- This's a nonhomogeneous second-order differential equation. Its complementary equation is the same as in the source-free parallel *RLC* circuit
- For the particular solution, we can easily find that a particular solution is $i = I_s$. The general solution is then given by

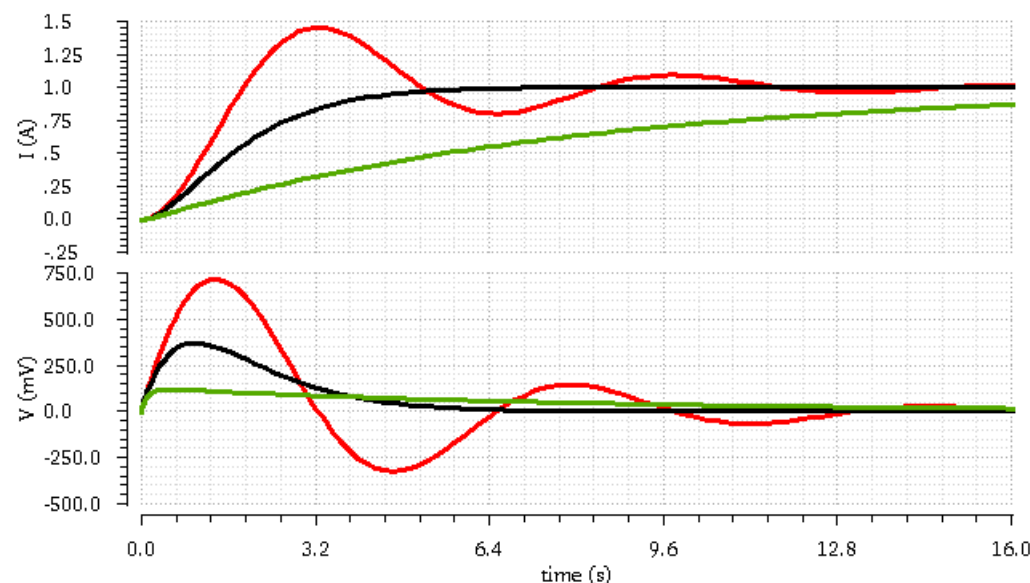
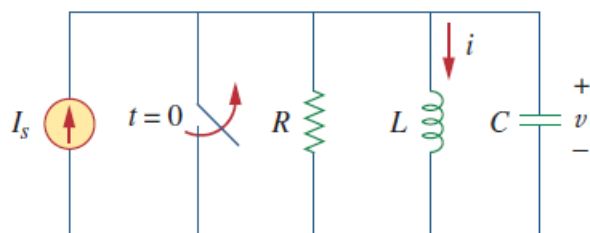
$$\begin{aligned} i(t) &= I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} && \text{(Overdamped)} \\ i(t) &= I_s + (A_1 + A_2 t) e^{-\alpha t} && \text{(Critically damped)} \\ i(t) &= I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} && \text{(Underdamped)} \end{aligned}$$

Step Response of a Parallel RLC Circuit (II)

- Notice that the particular solution is also the steady-state value of the inductor current: $i_{ss}(t) = i(\infty) = I_s$. The general solution can also be expressed as the sum of two parts:

$$i(t) = i_t(t) + i_{ss}(t)$$

- where $i_{ss}(t)$ is the steady-state part (i.e., the particular solution), while the $i_t(t)$ is the transient part (i.e., the part achieved by solving the complementary equation)



$$\alpha = \frac{1}{2RC} \quad \text{and} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Example:

$$L = 1 \text{ H}, C = 1 \text{ F}, I_0 = 0 \text{ A}, V_0 = 0 \text{ V}$$

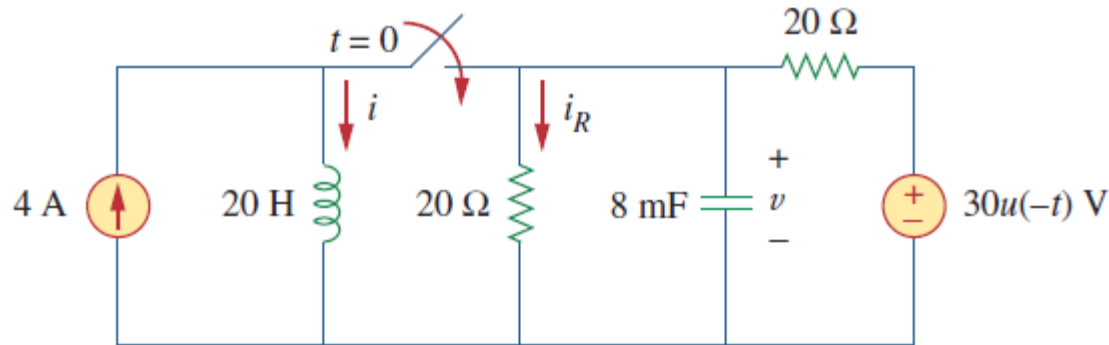
$G=0.5 \text{ S}, \alpha=0.25$, underdamped

$G=2 \text{ S}, \alpha=1$, critically damped

$G=8 \text{ S}, \alpha=4$, overdamped

Step Response of a Parallel *RLC* Circuit (I)

- Q: In the circuit below, find $i(t)$ and $i_R(t)$ for $t > 0$



Solution:

For $t < 0$, the 4-A current goes through the inductor which is short for dc. So, $i(0^+) = i(0^-) = I_0 = 4 \text{ A}$

The $30u(-t)$ is 30 V for $t < 0$, so

$$v(0^+) = v(0) = V_0 = 30 \cdot 20 / (20 + 20) = 15 \text{ V}$$

For $t > 0$, $30u(-t)$ is 0 V; and the switch is closed, so the circuit becomes a parallel *RLC* circuit supplied by a dc current source, with the two 20-Ω resistors in parallel as well, i.e., $R = 20 \parallel 20 = 10 \text{ Ω}$

The initial derivative of inductor current is $di(0^+)/dt = v(0^+)/L = 0.75 \text{ A/s}$

The steady-state inductor current is $i(\infty) = 4 \text{ A}$

The resonant frequency and damping factor are:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20 \times 8 \times 10^{-3}}} = 2.5, \quad \alpha = \frac{1}{2RC} = \frac{1}{2 \times 10 \times 8 \times 10^{-3}} = 6.25 > \omega_0$$

Step Response of a Parallel *RLC* Circuit (II)

So, the step response is over damped and given by

$$i(t) = 4 + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -0.5218, s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -11.978$$

Using the initial conditions, we get

$$\begin{cases} 4 = 4 + (A_1 + A_2) \\ 0.75 = (-0.5218A_1 \cdot 1 - 11.978A_2 \cdot 1) \end{cases} \rightarrow A_1 = 0.0655, A_2 = -0.0655$$

Hence, the response $i(t)$ is exactly given by

$$i(t) = 4 + 0.0655(e^{-0.5218t} - e^{-11.978t})$$

From $i(t)$, we can find other variables. The resistor current $i_R(t)$ is given by

$$i_R(t) = \frac{v(t)}{20} = L \frac{di}{dt} \frac{1}{20} = 0.785e^{-11.978t} - 0.0342e^{-0.5218t} \text{ A}$$

Procedure to Solving General 2nd-Order Circuits

- The approach to solving the series and parallel forms of *RLC* circuits can also be applied to **general second-order circuits** (一般二阶电路) with or without constant independent sources
- **Procedures:**
 1. Calculate the initial values: $x(0^+)$ and $x'(0^+)$
 2. Applying KCL and KVL to establish the second order differential equation. Find out the characteristic roots of its complementary equation (**homogeneous equation**). Write out the general solution or **transient solution** $x_t(t)$ with unknown constants to be determined
 3. Find out the **stead-state response** $x_{ss}(t) = x(\infty)$. The particular solution $x_p(t) = x(\infty)$
 4. The total response is given by $x(t) = x_p(t) + x_t(t)$. We finally determine the constants associated with the transient solution $x_t(t)$ by imposing the initial values $x(0^+)$ and $x'(0^+)$

Example: General Second Order Circuits

- Find the complete response $v(t)$ and $i(t)$ for $t > 0$ in the circuit below

Solution:

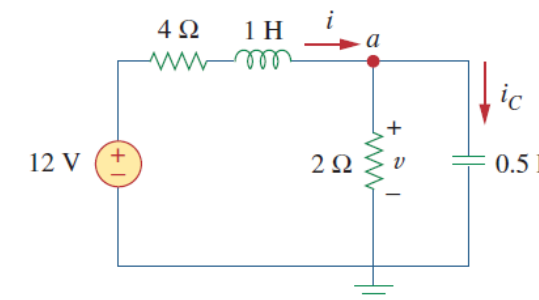
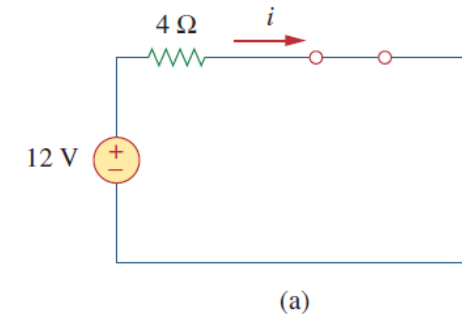
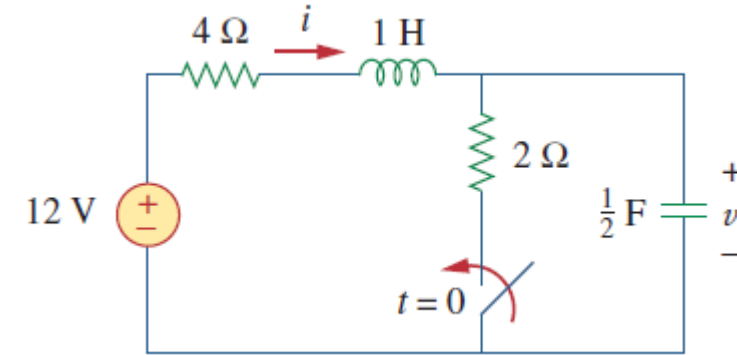
- Find the initial conditions: $v(0^+) = 12$ V,
 $v'(0) = i_C(0^+)/C = -6/0.5 = -12$ V/s.
- Applying KCL at the top right node and KVL at the outermost loop, we have

$$\begin{cases} i = \frac{v}{2} + C \frac{dv}{dt} = \frac{v}{2} + \frac{1}{2} \frac{dv}{dt} \\ 12 = 4i + L \frac{di}{dt} + v = 4i + \frac{di}{dt} + v \end{cases} \rightarrow \frac{d^2v}{dt^2} + 5 \frac{dv}{dt} + 6v = 24 \rightarrow s_1 = -2, s_2 = -3$$

So the transient solution is $v_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} = A_1 e^{-2t} + A_2 e^{-3t}$

- The final state is 4 V,
- The total solution $v(t) = 4 + A_1 e^{-2t} + A_2 e^{-3t} = 4 + 12e^{-2t} - 4e^{-3t}$ V

The inductor current $i(t) = \frac{v}{2} + \frac{1}{2} \frac{dv}{dt} = 2 - 6e^{-2t} + 4e^{-3t}$ A



$$\begin{aligned} i(0^+) &= i_C(0^+) + \frac{v(0^+)}{2} \\ 0 &= i_C(0^+) + \frac{12}{2} \\ i_C(0^+) &= -6 \text{ A} \end{aligned}$$

Second-Order Op Amp Circuits (I)

- The analysis of a second-order op amp circuit follows the same four steps procedures described in the previous sections
- Q: In the op amp circuit below, find $v_o(t)$ for $t > 0$ when $v_s(t) = 10u(t)$ mV. Let $R_1 = R_2 = 10 \text{ k}\Omega$, $C_1 = 20 \text{ }\mu\text{F}$, and $C_2 = 100 \text{ }\mu\text{F}$

Solution

The op amp forms a voltage follower

Therefore, the voltage across C_1 is the same as v_o

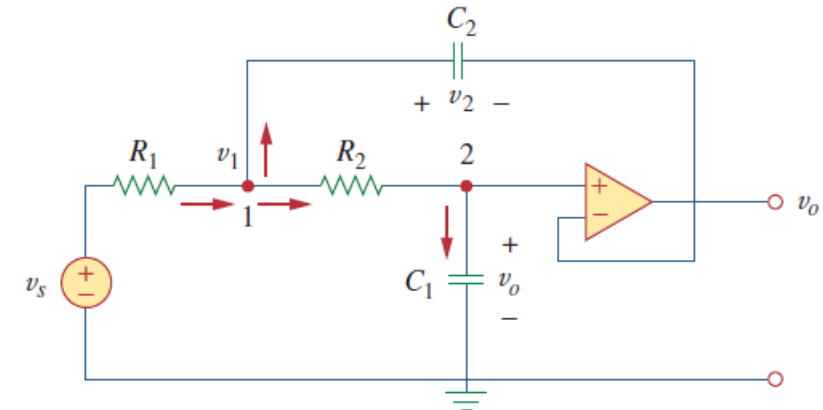
1. Initial values:

$$v(0) = 0;$$

$$v'(0) = (v_1(0) - v_o(0))/R_2 C_1 = 0 \text{ V/s}$$

2. Applying KCL to nodes 1 and 2, respectively, we can get

$$\begin{cases} \frac{v_s - v_1}{R_1} = C_2 \frac{d(v_1 - v_o)}{dt} + \frac{v_1 - v_o}{R_2} \\ \frac{v_1 - v_o}{R_2} = C_1 \frac{dv_o}{dt} \end{cases} \quad \Rightarrow \quad \frac{d^2 v_o}{dt^2} + \left(\frac{1}{R_2 C_2} + \frac{1}{R_1 C_2} \right) \frac{dv_o}{dt} + \frac{v_o}{R_1 R_2 C_1 C_2} = \frac{v_s}{R_1 R_2 C_1 C_2}$$



Second-Order Op Amp Circuits (II)

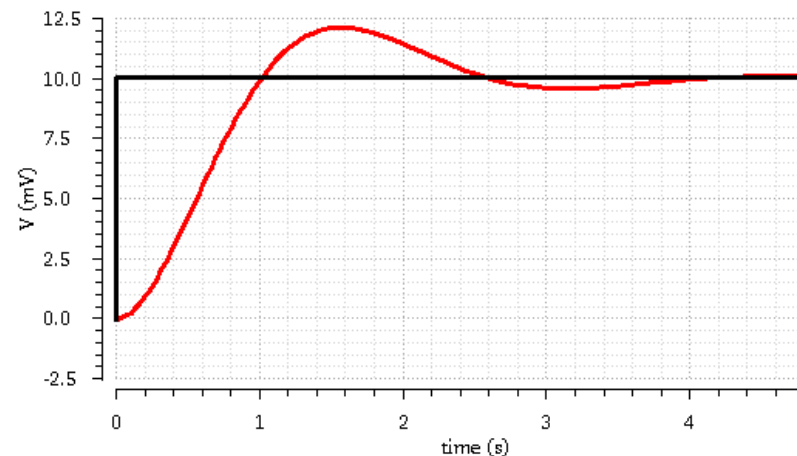
Substituting in the values of R_1 , R_2 , C_1 , C_2 , we have $v_o''(t) + 2v_o'(t) + 5v_o(t) = 5v_s$. The characteristic roots of this nonhomogeneous differential equation are $s_{1,2} = -1 \pm j2$. Hence the transient response is written by

$$v_{ot}(t) = e^{-t} (A_1 \cos 2t + A_2 \sin 2t)$$

3. The steady-state value $v_{oss}(t) = v_o(\infty) = 10$ mV.

4. The complete response $v_o(t) = v_{oss}(t) + v_{ot}(t) = 10 + e^{-t} (A_1 \cos 2t + A_2 \sin 2t)$

Taking account of the initial values, we finally have $v_o(t) = 10 + e^{-t} (-10 \cos 2t - 5 \sin 2t)$ mV



Summary

- The determination of the initial values $x(0^+)$ and $x'(0^+)$ and final value $x(\infty)$ is crucial to analyzing second-order circuits
- The RLC circuit are described by a second-order differential equation, whose solution can be written as the sum of a general solution (transient response) and a particular solution (steady response). The transient response can take three forms depending the relationship of the damping factor α and the resonant frequency $\omega_0 = 1/\sqrt{LC}$. For a series circuit $\alpha = R/2L$, and for a parallel circuit $\alpha = 1/2RC$
- If there are no independent sources in the circuit after switching, the response of the circuit to its initial stored energy is the natural response
- The natural response of an RLC circuit may be overdamped, critically damped or underdamped as α decreasing from a value larger than ω_0 to one smaller than ω_0 . The response is of critically damped when the roots are identical ($s_1 = s_2$ or $\alpha = \omega_0$), or overdamped when the roots are unequal real numbers ($\alpha > \omega_0$), or underdamped when the roots are complex conjugate ($s_1 = s_2^*$ or $\alpha < \omega_0$)
- If independent sources are present in the circuit after switching, the complete response is the sum of the transient response and the steady-state response